

Portfolio Tail Risk Analytics

Quantitative Analysis of Portfolio Tail Risk

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Abstract

This study evaluates the accuracy of four risk models—Historical Simulation, Normal Parametric, Student-t, and Monte Carlo Bootstrap—in estimating Value-at-Risk and Expected Shortfall across five multi-asset portfolios over the 2016–2025 period. The hypothesis that heavy-tailed models outperform the Normal distribution is supported by the results. The Normal model consistently fails to capture extreme losses, resulting in excessive VaR breaches during the 2022 market stress especially, while the Student-t model demonstrates greater robustness at the 99% confidence level. Expanding-window backtesting shows that all models perform adequately during calm market conditions but become less reliable during adverse market conditions such as the 2022 equity-bond correlation breakdown, when the correlation between VOO and TLT shifted from -0.38 to +0.08, weakening diversification benefits. Multi-scenario stress testing further shows that simultaneous declines across asset classes produce the most severe losses. The findings suggest that no single model is capable of fully capturing portfolio tail behaviour. Hence, a combined approach incorporating heavy-tailed distributions, backtesting, and scenario analysis would ideally provide a more robust and forward-looking risk assessment framework.

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Introduction

1.1 Background and Motivation

Investors run the risk of large losses, especially in times of market downturns. This tail risk is commonly quantified by **value-at-risk (VaR)** and **expected shortfall (ES)**. However, the assumed return distribution strongly influences the accuracy of these measures. Financial returns have fat tails, so the Normal distribution tends to underestimate the probability of large losses. The project compares different risk models, namely Historical, Normal, Student-t and Monte Carlo Bootstrap to establish the most reliable estimates for extreme losses. The models are then tested by backtesting and stress scenarios to determine how accurate they deal with extreme market stress.

1.2 Project Aim

The aim of this project is to assess the performance of different risk models in measuring portfolio tail risk and to identify the most accurate models for estimating extreme losses. This is done by comparing the models approaches to estimate VaR and Expected Shortfall. Throughout the project, various portfolios will be analysed, with the **Equal Weight** portfolio as the main example. The models are validated on an out-of-sample basis, through backtesting and stress scenarios, to assess the accuracy of the models under normal and stressed market conditions.

1.3 Research Hypothesis

The hypothesis of this project is that parametric models which account for heavier-tailed return distributions, such as the Student-t model, will provide more accurate estimates of extreme portfolio losses than models based on the Normal distribution, particularly during periods of market stress.

This hypothesis is tested by comparing the risk estimates produced by the different models and evaluating their out-of-sample performance through backtesting and stress testing. Particular attention is given to the performance of the models during the 2022 market stress period.

Data & Methodology

2.1 Data

The analysis uses daily adjusted closing prices for five assets: **VOO** (US Equities), **EFA** (International Equities), **TLT** (Long-Term US Treasuries), **AGG** (US Aggregate Bonds), and **GLD** (Gold). The data was sourced from Yahoo Finance via the yfinance library and covers the period **1 January 2016 to 31 December 2025**, providing approximately ten years of daily market data.

Adjusted closing prices were used since they incorporate dividends, stock splits and other corporate actions, giving a more accurate picture of total returns. The log returns were calculated from the adjusted prices by applying the formula:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Log returns are time additive and hence suitable for multi-period analysis. This makes log returns suitable for this project. Missing values were removed and the final dataset was saved as CSV formats for reproducibility. Clean and processed data is the basis for all subsequent phases like portfolio construction, risk measurement, backtesting and stress testing.

The selection of assets offers exposure to equities, fixed income and commodities, allowing the behaviour of diversified portfolios to be studied in different market environments.

2.2 Portfolio Construction

Five portfolio strategies were constructed using the five assets to reflect different investment styles and risk profiles. All portfolios are fully invested with a sum of 100% in weights. The weights remain constant over the period of analysis, meaning that the portfolios are rebalanced daily to ensure that the target allocations are maintained.

The five strategies are:

- **Equal Weight** – all five assets are given equal 20% weightings.
- **Classic 60/40** – a traditional balanced portfolio with 60% allocated to equities and 40% to bonds, with no gold exposure.

- **Defensive** – a low-risk portfolio heavily weighted toward bonds and gold, with only small equity exposure.
- **Aggressive Growth** – a high-risk portfolio dominated by equities, with a small bond allocation and no gold.
- **Conservative Income** – a bond-heavy portfolio designed for income and capital preservation, with moderate gold exposure.

The portfolio weights are summarised in the table below:

<i>Portfolio</i>	<i>VOO</i>	<i>EFA</i>	<i>TLT</i>	<i>AGG</i>	<i>GLD</i>
<i>Equal Weight</i>	20%	20%	20%	20%	20%
<i>Classic 60/40</i>	30%	30%	20%	20%	0%
<i>Defensive</i>	5%	5%	30%	30%	30%
<i>Aggressive Growth</i>	40%	40%	10%	10%	0%
<i>Conservative Income</i>	10%	10%	35%	35%	10%

Portfolio returns were calculated using the weighted sum of individual asset log returns, with the log return of a portfolio approximated as:

$$r_{p,t} = \ln \left(\sum_{i=1}^n w_i e^{r_{i,t}} \right)$$

where w_i is the weight of asset i , and $r_{i,t}$ is the log return of asset i at time t . This formula assumes daily rebalancing and provides an accurate representation of portfolio performance.

The main portfolio for detailed analysis across all phases is the **Equal Weight** portfolio, and the other four strategies are used as benchmarks for comparison. This enables the investigation of the impact of different asset allocations on tail risk under a common setting.

2.3 Basic Performance Metrics

The table below summarises the key performance metrics for all five portfolios over the period 2016–2025:

<i>Metric</i>	<i>Equal Weight</i>	<i>Classic 60/40</i>	<i>Defensive</i>	<i>Aggressive Growth</i>	<i>Conservative Income</i>
<i>Annual Drift</i>	8.16%	7.58%	6.12%	9.48%	4.63%
<i>Volatility</i>	8.80%	10.63%	8.59%	13.60%	8.05%
<i>Sharpe</i>	0.59	0.43	0.36	0.48	0.20
<i>Max DD (Loss)</i>	22.33%	24.86%	24.91%	26.35%	24.81%
<i>Total Return</i>	125.58%	112.86%	84.07%	157.31%	58.65%
<i>CAGR</i>	8.50%	7.87%	6.31%	9.95%	4.74%
<i>Skewness</i>	-0.51	-0.74	-0.22	-0.84	-0.48
<i>Kurtosis</i>	15.28	17.54	6.83	19.51	9.90

The **Equal Weight** portfolio yielded an annualised drift of 8.16% with a volatility of 8.80%, thus producing a Sharpe ratio of 0.59. The total return for the period was 125.58%, or a compound annual growth rate (CAGR) of 8.50%. The portfolio's maximum drawdown was 22.33% which means that at its worst point, it had lost just over a fifth of its value from its previous peak. The return distribution is slightly negative skewed, with a skewness of -0.51, and heavily fat-tailed, with a kurtosis of 15.28. Hence, indicating that large negative returns occur more frequently than a Normal distribution would imply. This is an important observation and supports the use of heavy-tailed models later in the analysis.

Note, the **risk-free rate is 3%**, which is the average yield of the 3-month US Treasury bill over the period. This is from FRED (Federal Reserve Economic Data), the popular economic database maintained by the Federal Reserve Bank of St. Louis.

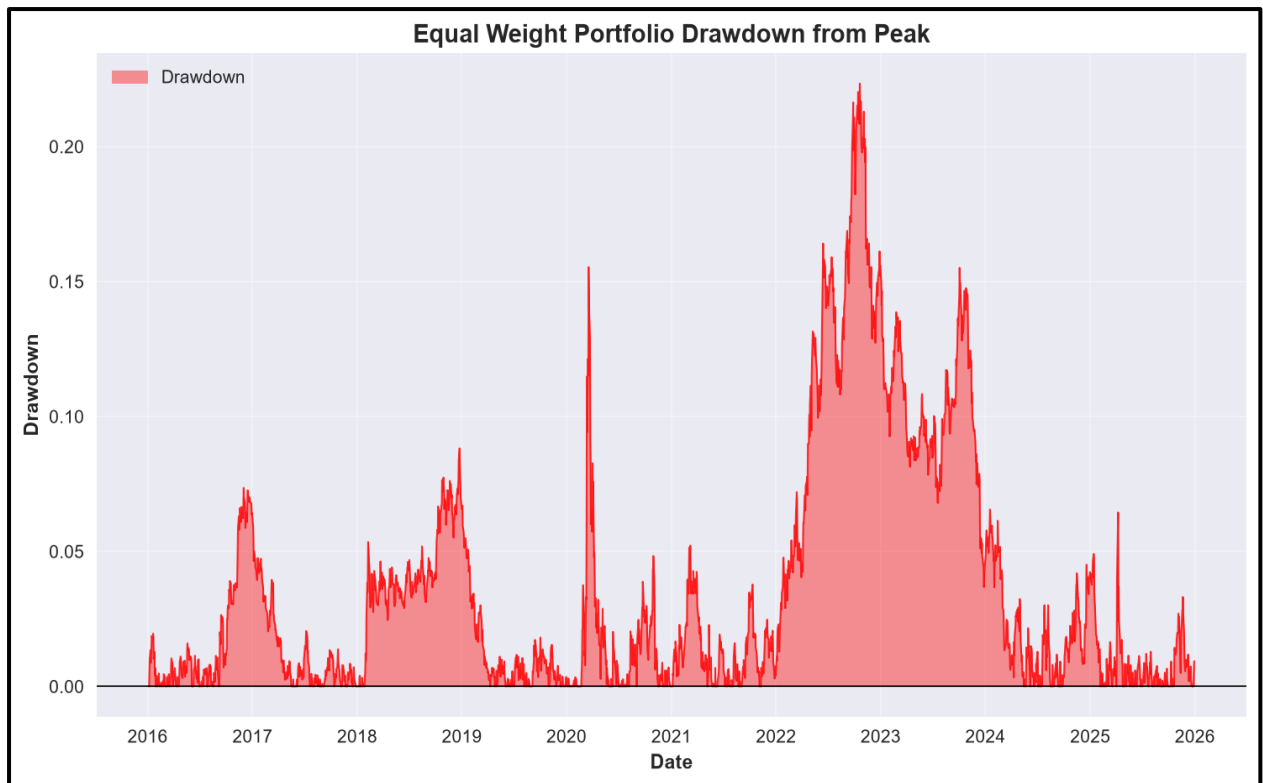


Figure 2.1



Figure 2.2

Figure 2.1 illustrates the drawdown of the **Equal Weight** portfolio over time. The chart illustrates the way in which losses pile up during market declines, and the length of time it takes for the portfolio to regain previous peaks. The biggest drawdown of 22.33% is

clearly visible and demonstrates the realistic effect of tail risk beyond standard measures of volatility.

Growth of all five portfolios is shown in **Figure 2.2**. **Aggressive Growth** had the best long-term growth, exceeding 157%, but also the worst volatility and the deepest drawdown of 26.35%. Conversely, **Conservative Income** had the lowest volatility but also the lowest total return equalling 58.65%, piling in comparison to the **Aggressive Growth** portfolio. **Equal Weight** is somewhere in between, with a balanced risk-return profile.

All portfolios display negative skewness and kurtosis well above 3, confirming that fat tails and asymmetric downside risk are present regardless of asset allocation. This motivates the detailed tail-risk analysis in the following sections.

Risk Model Estimation

3.1 Historical Simulation

Historical simulation estimates tail risk by looking at the actual distribution of past returns. VaR is set at the 5th (95%) and 1st (99%) percentiles of daily log returns, and ES is the average loss beyond those points. All results are shown as positive loss percentages:

Metric	Equal Weight	Classic 60/40	Defensive	Aggressive Growth	Conservative Income
95% VaR	0.82%	0.98%	0.84%	1.24%	0.82%
95% ES	1.27%	1.58%	1.21%	2.03%	1.17%
99% VaR	1.36%	1.72%	1.40%	2.28%	1.36%
99% ES	2.12%	2.84%	1.87%	3.73%	1.84%

For the **Equal Weight** portfolio on a **€100,000** position:

- 95% VaR: €820
- 99% VaR: €1,358
- 95% ES: €1,270
- 99% ES: €2,123

The 99% ES indicates that, on average, losses exceeded €2,000 in the worst 1% of days. ES is consistently larger than VaR, reflecting the severity of extreme tail losses.

Aggressive Growth shows the highest tail risk, while **Conservative Income** and **Defensive** show the lowest, consistent with their bond-heavy allocations.

Figure 3.1 below displays the daily log returns for the Equal Weight portfolio, with VaR and ES thresholds marked. Dashed lines show VaR at 95% and 99%, and dotted lines show ES. The histogram makes it clear that there are more extreme negative returns than a Normal distribution would predict. This shows why it is important to use tail-focused measures instead of just standard deviation.

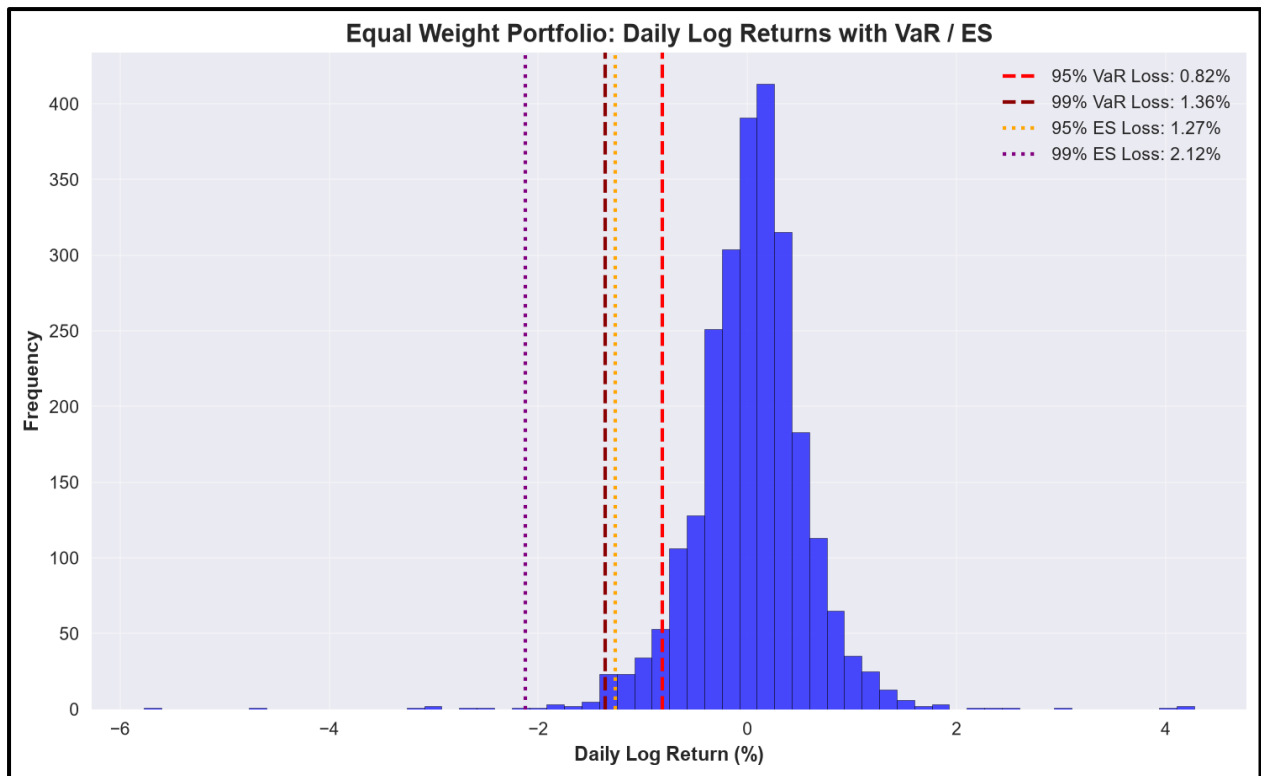


Figure 3.1

3.2 Normal Parametric VaR & ES

The Normal model assumes daily log returns follow a Gaussian distribution, described only by its mean and standard deviation. This makes the model simple to use, but it also means the tails shrink quickly, so extreme returns are seen as very rare contradicting real financial markets which are often victim to extreme shocks.

VaR is calculated directly from the fitted Normal distribution, while ES is derived using the truncated Normal formula. All values are reported as positive loss percentages.

Portfolio	95% VaR	95% ES	99% VaR	99% ES
<i>Equal Weight</i>	0.88%	1.11%	1.26%	1.45%
<i>Classic 60/40</i>	1.07%	1.35%	1.53%	1.75%
<i>Defensive</i>	0.87%	1.09%	1.23%	1.42%
<i>Aggressive Growth</i>	1.37%	1.73%	1.96%	2.25%
<i>Conservative Income</i>	0.82%	1.03%	1.16%	1.33%

For the **Equal Weight** portfolio, the Normal model has a 99% ES of 1.45% (€1,445), meaning the average loss in the worst 1% of days is estimated at just under €1,500.

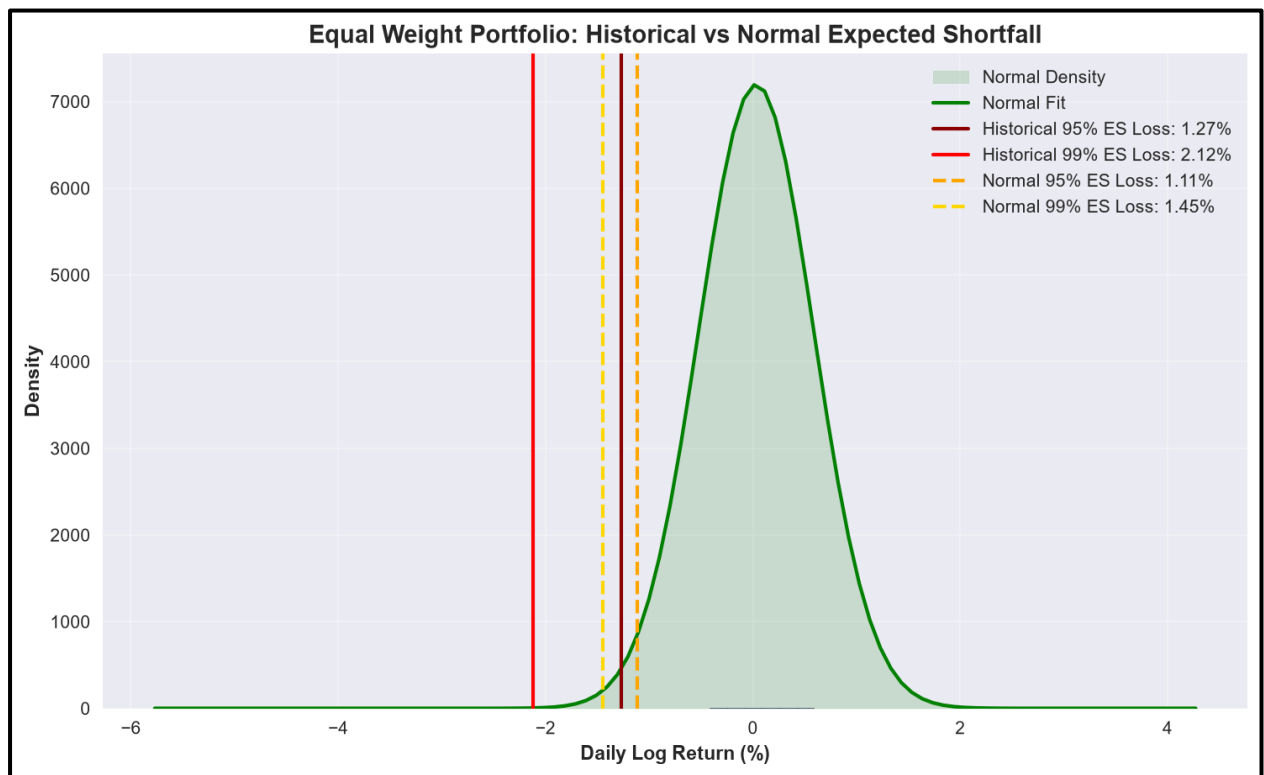


Figure 3.2

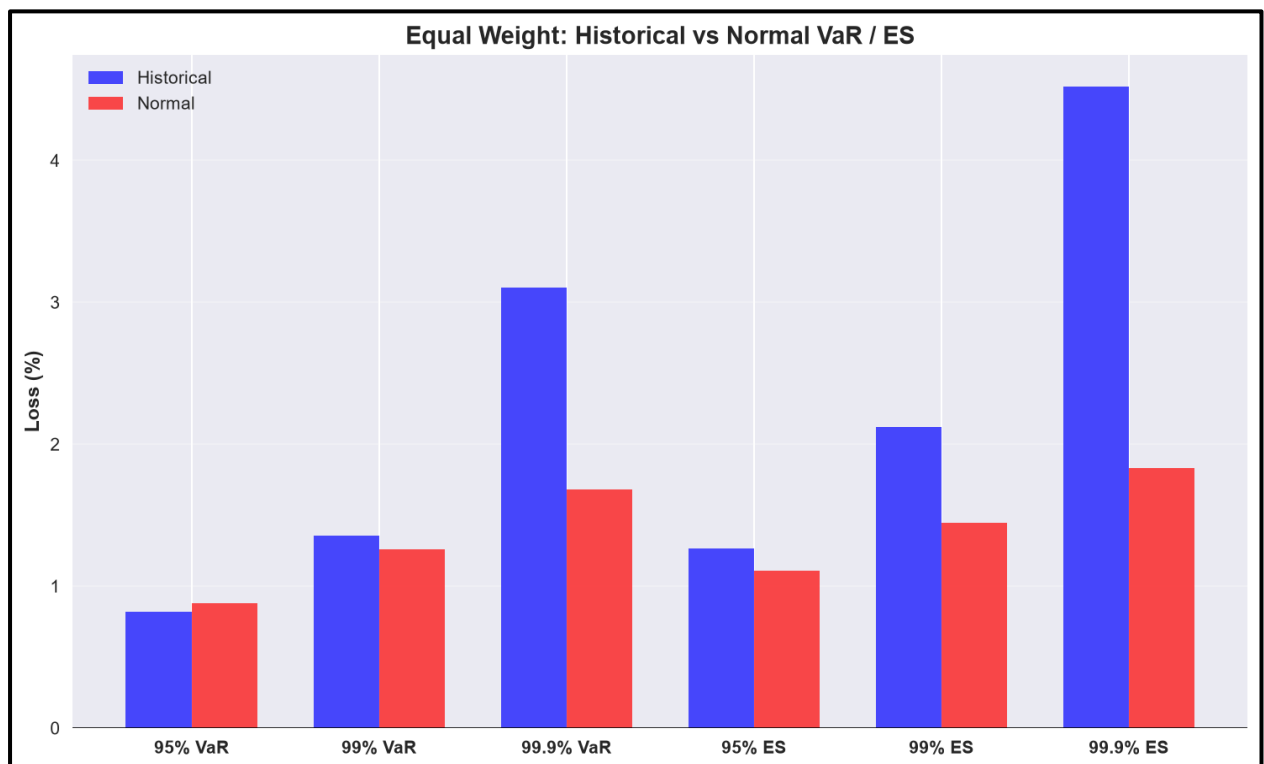


Figure 3.3

Figure 3.2 overlays the fitted Normal density on the **Equal Weight** return histogram, with Expected Shortfall thresholds marked at the 95% and 99% levels. The fit looks reasonable in the centre, but it visibly underestimates the tail.

Figure 3.3 compares VaR and ES for Historical and Normal approaches across three confidence levels: 95%, 99%, and 99.9%. The gap between the two models widens as we move further into the tail, especially for ES.

These results show that the Normal model underestimates tail risk, so a model with heavier tails is needed. The next section looks at the Student-t model.

3.3 Student-t VaR & ES

The Student-t model allows for heavier tails than the Normal model. It adds a parameter called degrees of freedom, which controls how fat the tails are. Lower degrees of freedom mean heavier tails and a higher chance of extreme outcomes.

<i>Portfolio</i>	<i>DF</i>	<i>95% VaR</i>	<i>95% ES</i>	<i>99% VaR</i>	<i>99% ES</i>
Equal Weight	3.89	0.77%	1.19%	1.40%	1.98%
Classic 60/40	1.98	1.06%	2.29%	2.58%	5.29%
Defensive	5.31	0.81%	1.17%	1.37%	1.80%
Aggressive Growth	3.05	1.14%	1.91%	2.25%	3.48%
Conservative Income	2.06	0.89%	1.85%	2.11%	4.15%

For the **Equal Weight** portfolio, the fitted degrees of freedom are 3.89, which suggests fat tails. The 99% VaR is 1.40% (€1,400), and the 99% ES is 1.98% (€1,984). Comparing the models:

<i>Metric</i>	<i>Historical</i>	<i>Normal</i>	<i>Student-t</i>
95% VaR	0.82%	0.88%	0.77%
99% VaR	1.36%	1.26%	1.40%
95% ES	1.27%	1.11%	1.19%
99% ES	2.12%	1.45%	1.98%

The Student-t model gives a 99% ES of 1.98%, much closer to the historical 2.12% than the Normal model's 1.45%. It still slightly understates the historical tail, but the improvement is substantial.

Figure 3.4 compares all three models' side by side for VaR and ES at the 95% and 99% confidence levels. The Student-t bars are noticeably closer to the historical bars than the Normal bars, especially for ES, due to Student-t's degrees of freedom parameter, allowing extreme losses to be captured.

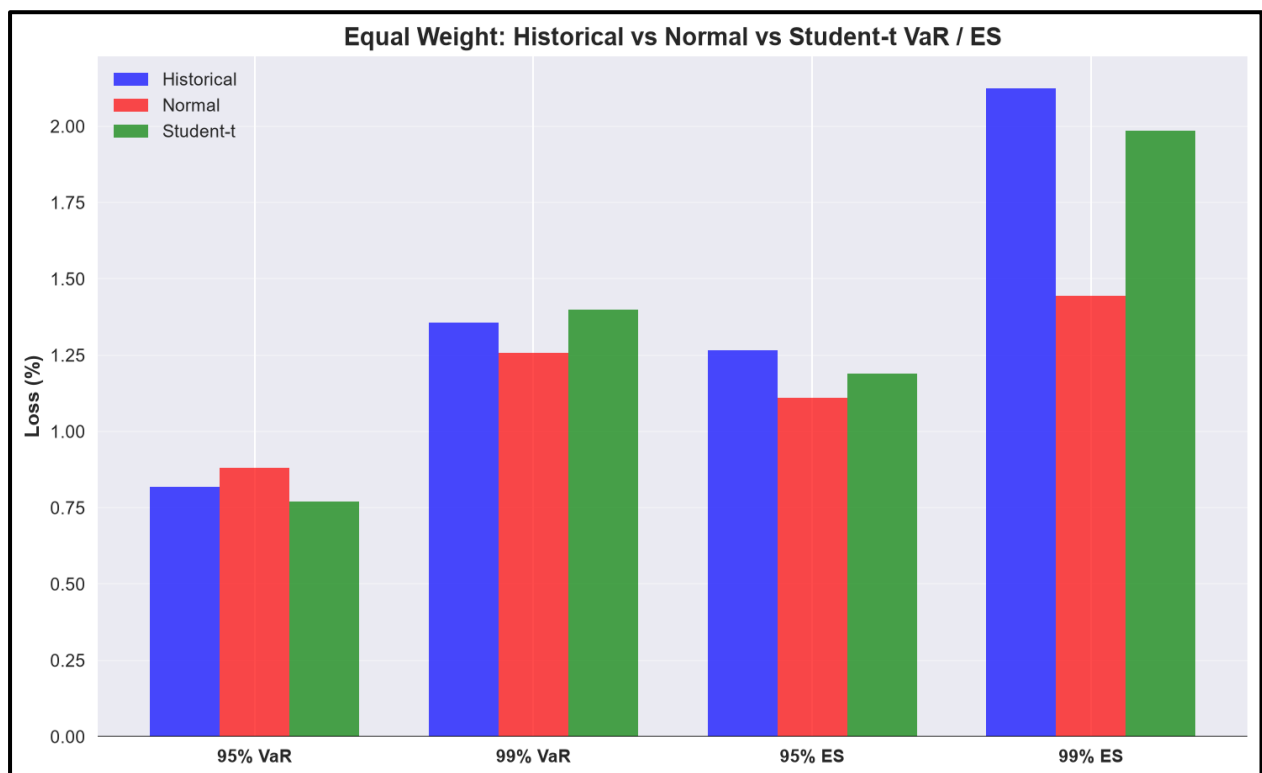


Figure 3.4

The results provide initial support for the hypothesis, a heavy-tailed model like the Student-t gives more conservative and accurate tail risk estimates than the Normal distribution.

3.4 Monte Carlo Bootstrap Simulation

The bootstrap approach creates a simulated log return distribution by resampling daily log returns from the historical data. This is repeated 10,000 times, with each simulation randomly picking one day from the historical sample. Because it uses actual log returns, the bootstrap does not assume any distribution shape and just uses the data as it is. The results are as follows:

Portfolio	95% VaR	95% ES	99% VaR	99% ES
Equal Weight	0.80%	1.22%	1.35%	1.96%
Classic 60/40	0.99%	1.67%	1.86%	3.23%
Defensive	0.84%	1.21%	1.40%	1.83%
Aggressive Growth	1.26%	1.99%	2.30%	3.44%
Conservative Income	0.82%	1.17%	1.32%	1.91%

For the **Equal Weight** portfolio, the bootstrap gives a 95% VaR of 0.80% and a 99% VaR of 1.35%. The 99% ES is 1.96%, which is close to the historical estimate of 2.12%.

This result is expected. Because the bootstrap uses the historical log return distribution directly, its tail estimates are very similar to the Historical Simulation results. Small differences are due to random sampling, but the overall shape of the tail stays the same since no assumptions are made about the distribution. Comparing all four models:

Metric	Historical	Normal	Student-t	Bootstrap
95% VaR	0.82%	0.88%	0.77%	0.80%
99% VaR	1.36%	1.26%	1.40%	1.35%
95% ES	1.27%	1.11%	1.19%	1.22%
99% ES	2.12%	1.45%	1.98%	1.96%

The bootstrap and Historical Simulation are almost identical over a singular day horizon. The Normal model still stands out as the outlier, especially at the 99% ES level, where it produces the lowest estimate by far.

In summary, the historical distribution is still the benchmark. The Normal model continues underestimating tail risk, while the Student-t and Monte Carlo Bootstrap models match the actual data more closely.

For visual reference, **Figure 3.5** compares VaR and ES across all four models simultaneously.

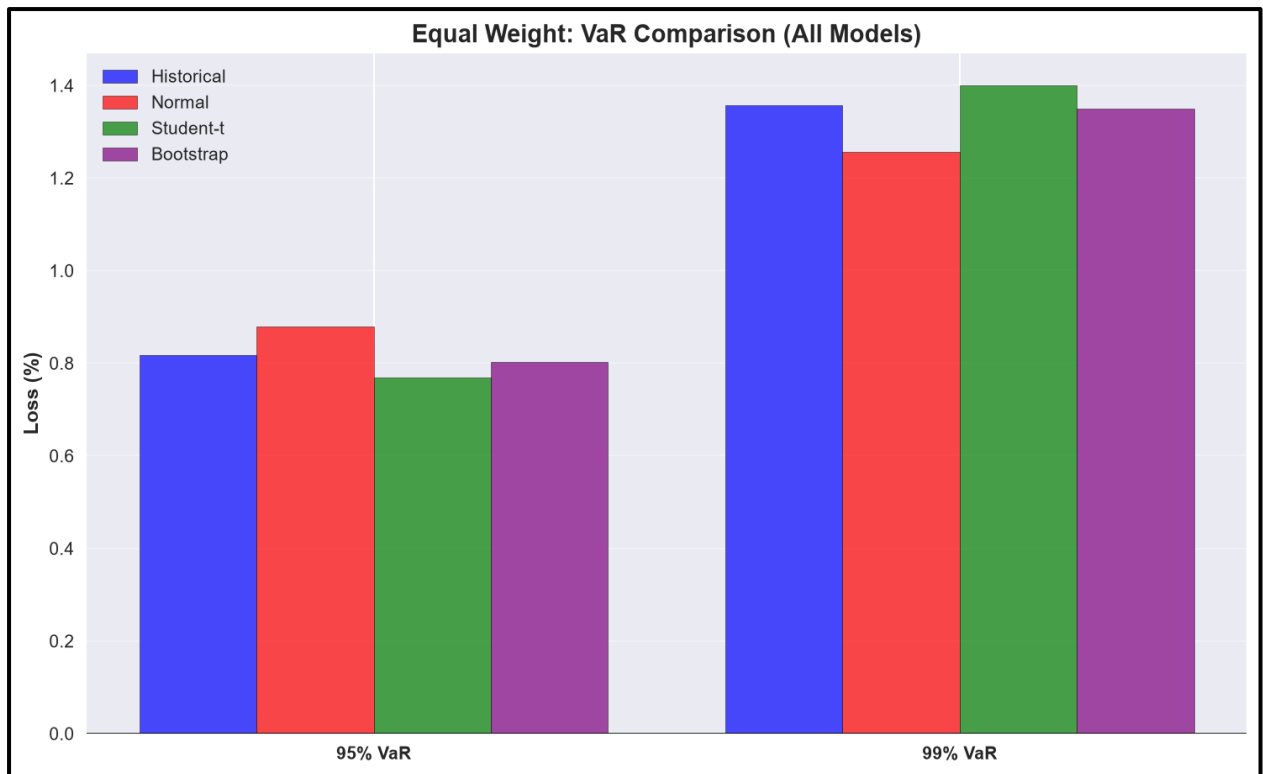


Figure 3.5

Expanding Window Backtesting

4.1 Backtesting Methodology

Backtesting tests how well a VaR model would have performed out-of-sample. For each year in the test period, only past data is used to estimate the model, and then the number of days where actual losses exceeded the predicted VaR is counted.

The backtest uses an expanding window approach. For each test year, the model is estimated using all data available up to that year. This reflects how risk models can be updated as new data comes in.

Three VaR models are tested at the 99% and 95% confidence levels:

- **Historical** – uses the empirical quantile of past returns.
- **Normal** – fits a Gaussian distribution.
- **Student-t** – fits a heavy-tailed distribution.

A breach happens when the actual loss on a day is bigger than the predicted VaR. For example, at the 99% level, we expect about 1 breach for every 100 trading days.

To check if the number of breaches is reasonable, the Kupiec test is used. It compares how often breaches occur with how often they are expected and gives a p-value which gives an insight into how accurate a model is.

The next section applies this framework and examines the results year by year.

4.2 Equal Weight Backtest Results

The table below shows the backtest results for the **Equal Weight** portfolio from 2021 to 2025. For each year, the number of breaches is shown for the three models at the 99% and 95% confidence levels, alongside the expected number:

<i>Confidence</i>	<i>Model</i>	<i>2021</i>	<i>2022</i>	<i>2023</i>	<i>2024</i>	<i>2025</i>
99%	Expected	2.5	2.5	2.5	2.5	2.5
	Historical	2	10	2	3	2
	Normal	3	19	3	4	5
	Student-t	3	2	0	0	1

Confidence	Model	2021	2022	2023	2024	2025
95%	Expected	12.6	12.6	12.5	12.6	12.5
	Historical	16	71	18	15	9
	Normal	10	44	15	12	8
	Student-t	15	47	15	11	7

Figure 4.1 below shows the 99% VaR breaches for the **Equal Weight** portfolio across the five test years. The bars compare the three models against the expected number of breaches.

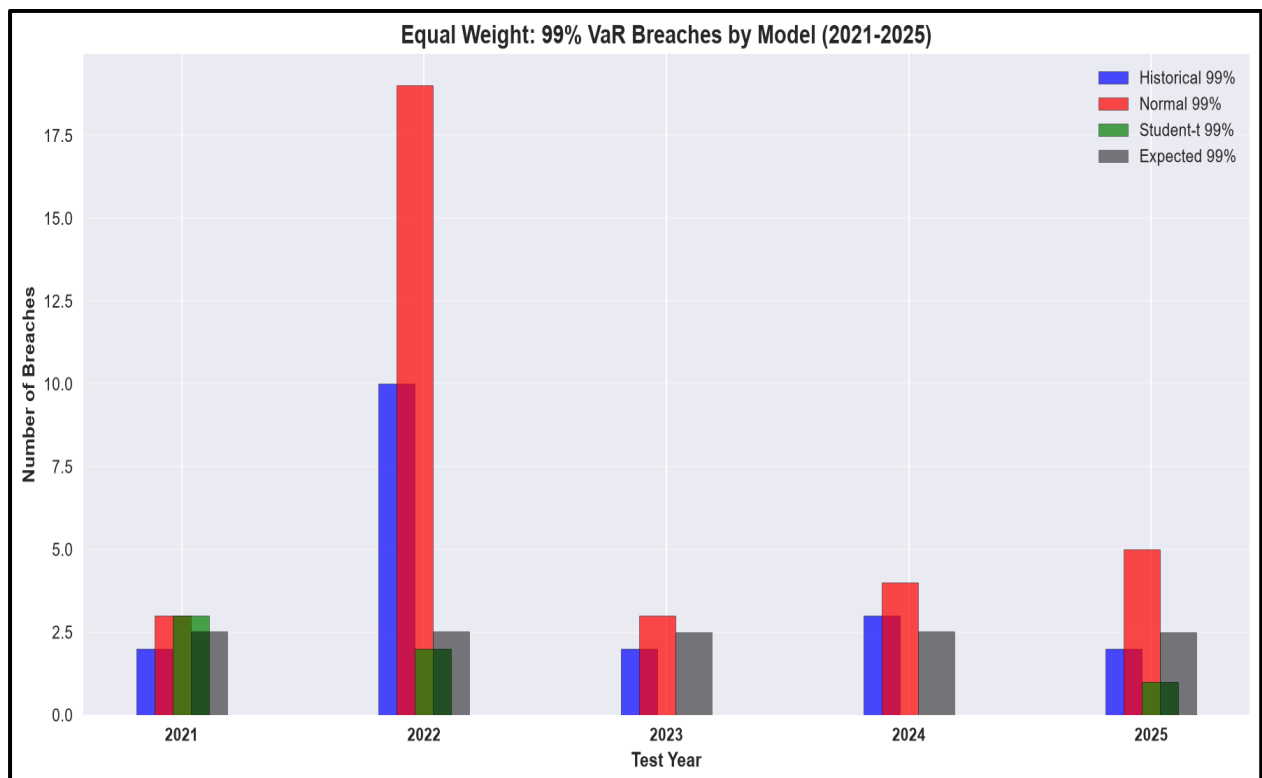


Figure 4.1

In most years, the number of breaches at the 99% level is close to the expected 2.5. The models perform similarly in 2021, 2023, 2024, and 2025, with only small differences from what is expected.

The clear exception is 2022. During that year, markets experienced a sharp simultaneous sell-off in equities and bonds. At the 99% level, the Historical model recorded 10 breaches and the Normal model 19 breaches, far above the expected 2.5.

The Student-t model, however, recorded only 2 breaches, staying close to expectations. Its fatter tails produced a higher VaR threshold that absorbed the extreme moves better than the other models.

At the 95% level, all three models failed in 2022. The Historical model had 71 breaches, Normal had 44, and Student-t 47. This shows that while the Student-t model was more robust at the very extreme tail, it still could not cope with the sheer number of moderately large losses during the crisis.

In 2023 and 2024, the Student-t model had 0 breaches at the 99% level. While this might look like excellent performance, it also suggests the model may be too conservative, setting a VaR threshold so high that it is rarely exceeded.

Overall, the **Equal Weight** results show two main patterns. First, in normal markets, all three models work well and have similar breach counts. Second, during market stress, the Normal model performs poorly, while the Student-t model is more resilient but still not perfect.

4.3 Kupiec Test

The Kupiec test checks whether the number of VaR breaches is statistically consistent with the confidence level used. For a 99% VaR, we expect about 1% of days to breach – roughly 2.5 days per year. For a 95% VaR, the expected number is around 12.6 days per year.

The test statistic compares the observed breach rate to the expected breach rate. The formula is:

$$LR = -2 \ln \left(\frac{(1-p)^{T-x} p^x}{\left(1 - \frac{x}{T}\right)^{T-x} \left(\frac{x}{T}\right)^x} \right)$$

where T is the number of trading days, x is the number of breaches, and p is the expected breach probability.

The p-value is then taken from a chi-squared distribution with one degree of freedom. There are then two results:

- **$p > 0.05$** → fail to reject the null hypothesis. The model is suitable.
- **$p < 0.05$** → reject the hypotheses. There is evidence that the model isn't suitable.

The results follow:

Year	Hist 99% p	Norm 99% p	t 99% p	Hist 95% p	Norm 95% p	t 95% p
2021	0.733	0.768	0.768	0.345	0.436	0.500
2022	0.000	0.000	0.737	0.000	0.000	0.000
2023	0.742	0.758	nan	0.133	0.481	0.481
2024	0.768	0.388	nan	0.500	0.861	0.637
2025	0.747	0.160	0.281	0.292	0.167	0.085

In normal years, most models pass with p-values above 0.05. The nan values for the Student-t model in 2023 and 2024 happen because the model had zero breaches. When this occurs, the likelihood ratio is undefined, so no p-value can be calculated. This does not mean the model passed or failed; it just cannot be tested in those years.

In 2022, the Historical and Normal models both produced p-values of 0.000 at both confidence levels, meaning they were decisively rejected. The Student-t model passed at the 99% level ($p = 0.737$) but failed at the 95% level ($p = 0.000$).

The Kupiec test confirms earlier observations. The Normal model is unreliable during market stress, while the Student-t model is more robust but yet has imperfections.

4.4 Other Portfolios in the Backtest

The backtest was also run for the other four portfolios to see how portfolio composition affects model performance. The table below shows the number of 99% VaR breaches in 2022, the most difficult year in the test period:

Portfolio	Historical 99%	Normal 99%	Student-t 99%	Expected
Equal Weight	10	19	2	2.5
Classic 60/40	7	14	9	2.5
Conservative Income	28	28	25	2.5
Defensive	15	17	1	2.5
Aggressive Growth	8	13	8	2.5

Figure 4.2 highlights the same data visually, comparing breaches across all five portfolios in 2022. The chart makes it clear that no model performed well across the board.

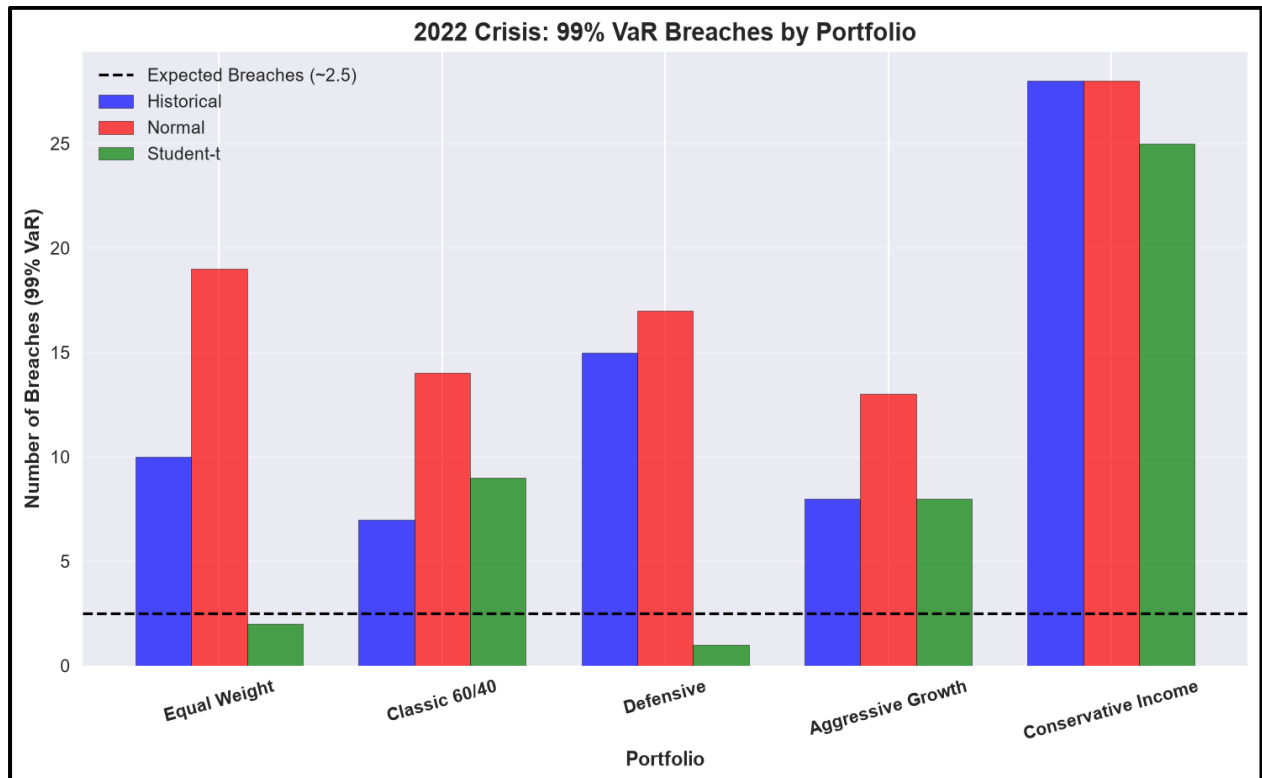


Figure 4.2

Every portfolio had breaches far above the expected 2.5, except for the Student-t model for the **Defensive** portfolio, which had just 1 breach. This matches the earlier **Equal Weight** result, where the Student-t model's fatter tails gave better protection against extreme losses.

The worst performer was **Conservative Income**, with 28 breaches under both the Historical and Normal models. This portfolio is heavily weighted toward bonds, which suffered unusually large losses in 2022 as interest rates rose sharply. The model had been estimated on a period when bonds were relatively stable, so it set a VaR threshold that was far too low for the new environment.

Aggressive Growth also struggled, with 8 breaches under Historical and 13 under Normal. Equity markets declined significantly in 2022, but the equity-heavy portfolios were somewhat more prepared because their VaR thresholds were already higher due to higher volatility.

The pattern across all portfolios confirms that 2022 was a systematic shock that affected every strategy, not just one portfolio type. The failure was not just due to poor model choice, but to a major change in market behaviour, especially the breakdown of

the negative correlation between equities and bonds. When both asset classes fell together, diversification offered little protection, and models based on historical relationships underestimated the size of possible losses.

The next chapter examines this correlation breakdown in more detail and uses stress testing to evaluate how the portfolios would perform under specific adverse scenarios.

Multi-Scenario Stress Testing

5.1 Scenario Definitions

Stress testing evaluates how portfolios would perform under future hypothetical but plausible market shocks. Unlike VaR, which relies on historical data, stress testing imposes specific adverse conditions and calculates the resulting portfolio loss.

Five scenarios were used, each based on real or hypothetical market events:

<i>Scenario</i>	<i>VOO</i>	<i>EFA</i>	<i>TLT</i>	<i>AGG</i>	<i>GLD</i>
<i>2008 Crisis</i>	-40%	-45%	+15%	+5%	+25%
<i>Geopolitical</i>	-15%	-20%	+5%	+5%	+30%
<i>Stagflation</i>	-20%	-25%	-20%	-10%	+35%
<i>Solvency II</i>	-40%	-40%	-25%	-10%	-15%
<i>Tech Bubble</i>	-45%	-35%	+10%	+5%	-10%

The scenarios capture different types of market stress:

- **2008 Crisis** – a global financial shock with equity sell-offs and flight to safe assets (bonds and gold rise).
- **Geopolitical** – a moderate risk-off event with equity losses but bonds and gold providing support.
- **Stagflation** – rising inflation and weak growth, hitting equities and bonds simultaneously while gold rallies.
- **Solvency II** – a severe regulatory stress scenario where nearly all assets decline, including gold.
- **Tech Bubble** – a sharp equity correction led by technology stocks, with bonds rising but gold falling.

These scenarios were chosen to test the portfolios against a range of adverse conditions, from moderate to extreme, and to see whether diversification holds up when correlations change. The next section presents the results for each portfolio.

5.2 Stress Test Results

The table below shows the loss percentages for each portfolio under the five stress scenarios. Positive values represent losses, while negative values indicate gains:

Scenario	Equal Weight	Classic 60/40	Defensive	Aggressive Growth	Conservative Income
2008 Crisis	8.0%	21.5%	-9.3%	32.0%	-1.0%
Geopolitical	-1.0%	8.5%	-10.3%	13.0%	-3.0%
Solvency II	26.0%	31.0%	19.0%	35.5%	21.8%
Stagflation	8.0%	19.5%	0.8%	21.0%	11.5%
Tech Bubble	15.0%	21.0%	2.5%	30.5%	3.8%

Figure 5.1 displays the same results graphically through a line chart.

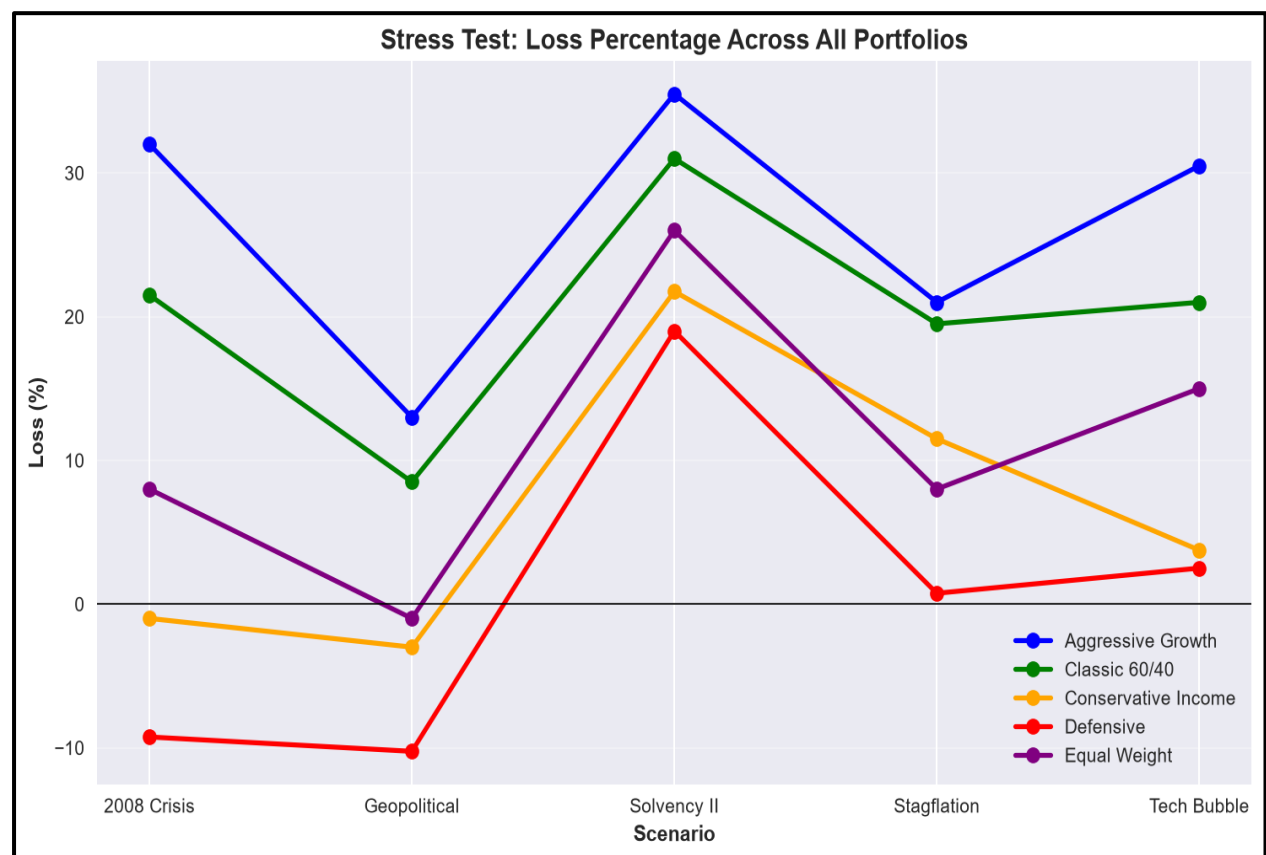


Figure 5.1

Differences across portfolios are evident in the line chart. **Aggressive Growth** is at the top in almost every scenario, consistently having the largest losses because of its heavy equity weighting and lack of defensive assets. In the **Solvency II** scenario, it loses 35.5% of its value, which is the worst result among all portfolios.

Equal Weight sits in the middle of the pack, with losses ranging from a small gain in the Geopolitical scenario to a loss of 26.0% under **Solvency II**. This reflects its balanced allocation across equities, bonds, and gold.

The **Defensive** portfolio stands out for its resilience. In three of the five scenarios, it makes a gain. For example, in the **2008 Crisis** scenario, it gains 9.3% because its heavy allocation to bonds and gold benefits from the flight to safety. Even in the worst-case **Solvency II** scenario, its loss is limited to 19.0%, the smallest among all portfolios.

Conservative Income also performs well in scenarios where bonds rally, but it struggles in **Stagflation** and **Solvency II**, where bonds decline. Its loss under **Solvency II** is 21.8%, larger than the **Defensive** portfolio's loss, reflecting its higher equity exposure and lower gold allocation.

Figure 5.2 presents the same results as a heatmap.

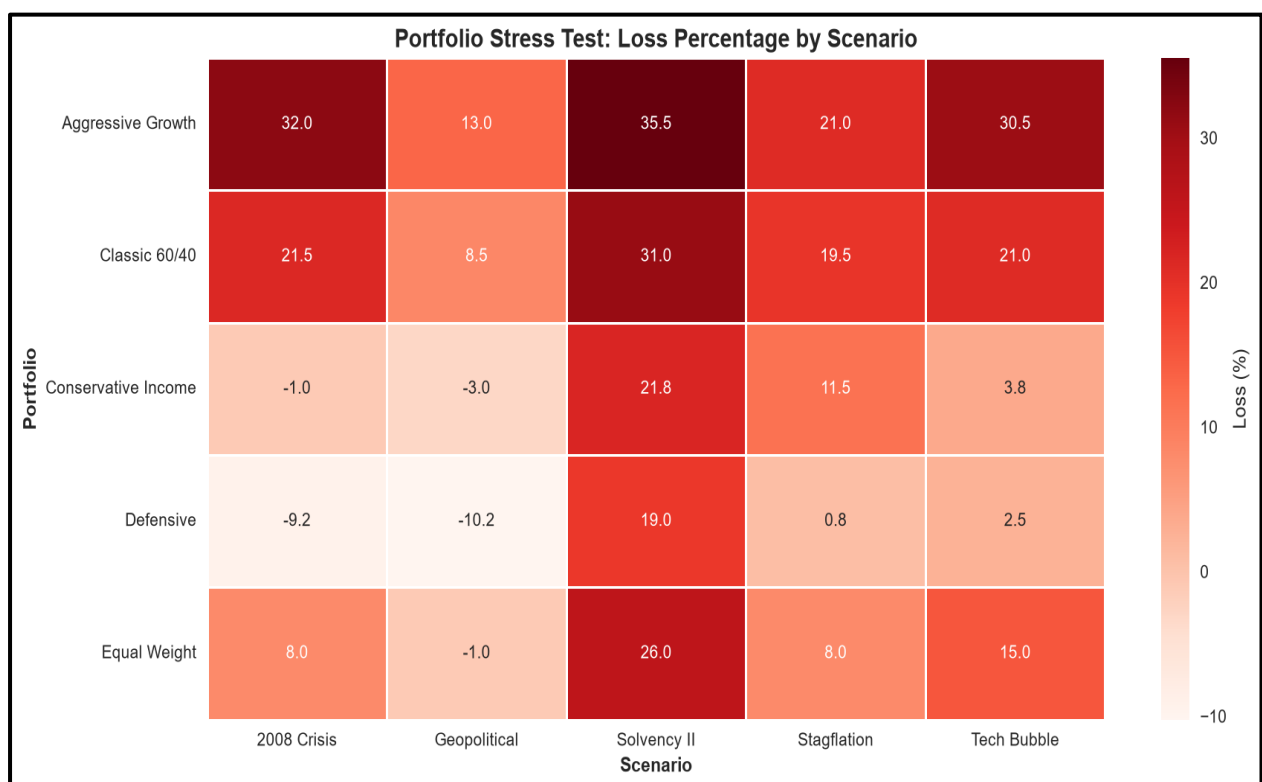


Figure 5.2

The heatmap shows the same patterns. The **Solvency II** column is the darkest for all portfolios, confirming it is the most severe scenario.

The results highlight how portfolio construction drives stress test outcomes. Portfolios with meaningful gold and bond allocations (**Defensive, Conservative Income**) are more resilient in most scenarios, while equity-heavy portfolios (**Aggressive Growth, Classic 60/40**) suffer the largest losses. The next section examines why this happens by looking at the correlations between assets.

5.3 Correlation Analysis

A key reason why some portfolios performed above expectations in certain scenarios previously is the changing relationship between assets, especially the correlation between equities and bonds.

Correlation measures how two assets move relative to each other. A correlation of +1 means they move in perfect tandem, while -1 means they move in opposite directions. Diversification works best when assets have low or negative correlations, because losses in one can be offset by gains in another.

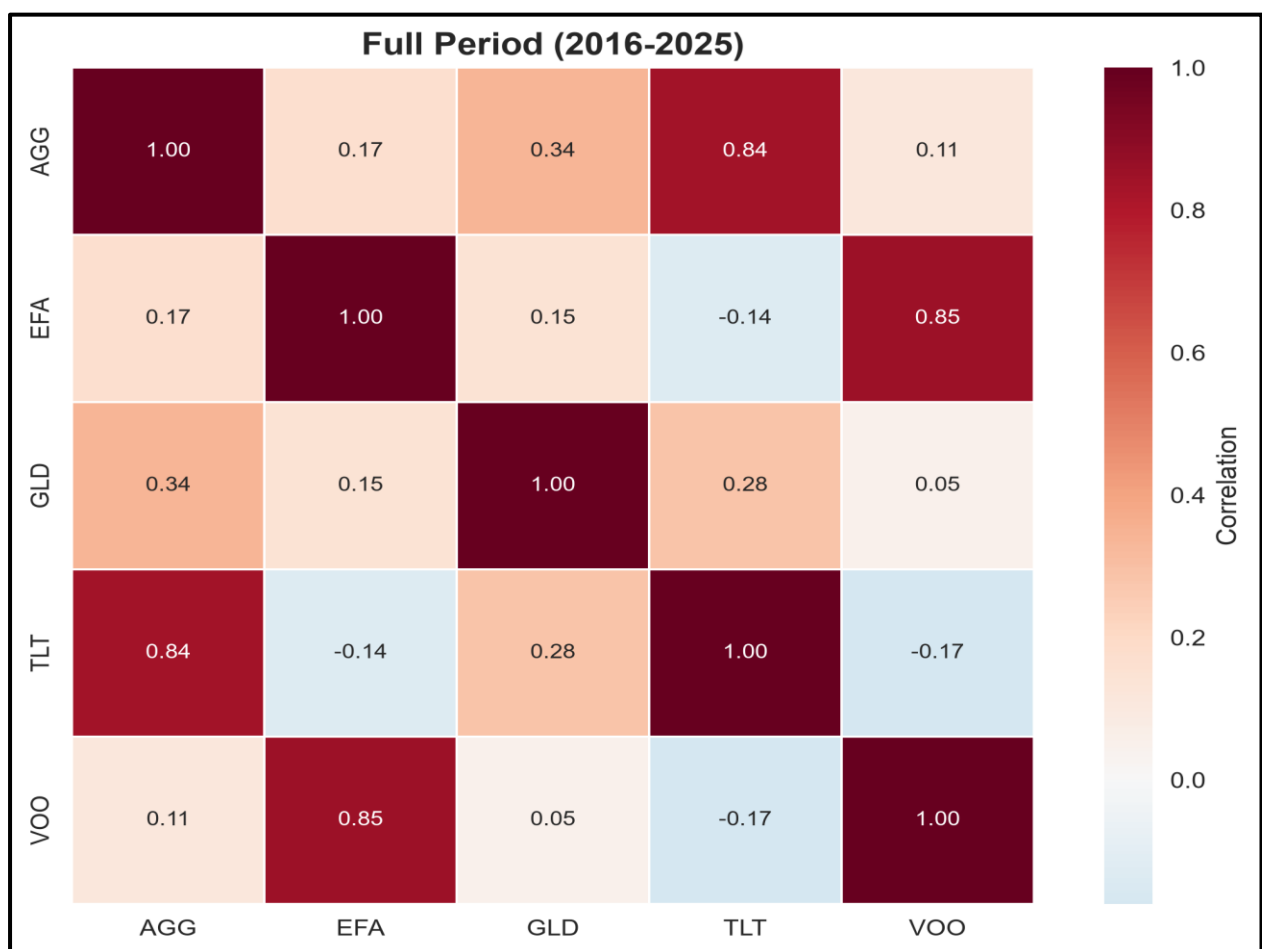


Figure 5.3

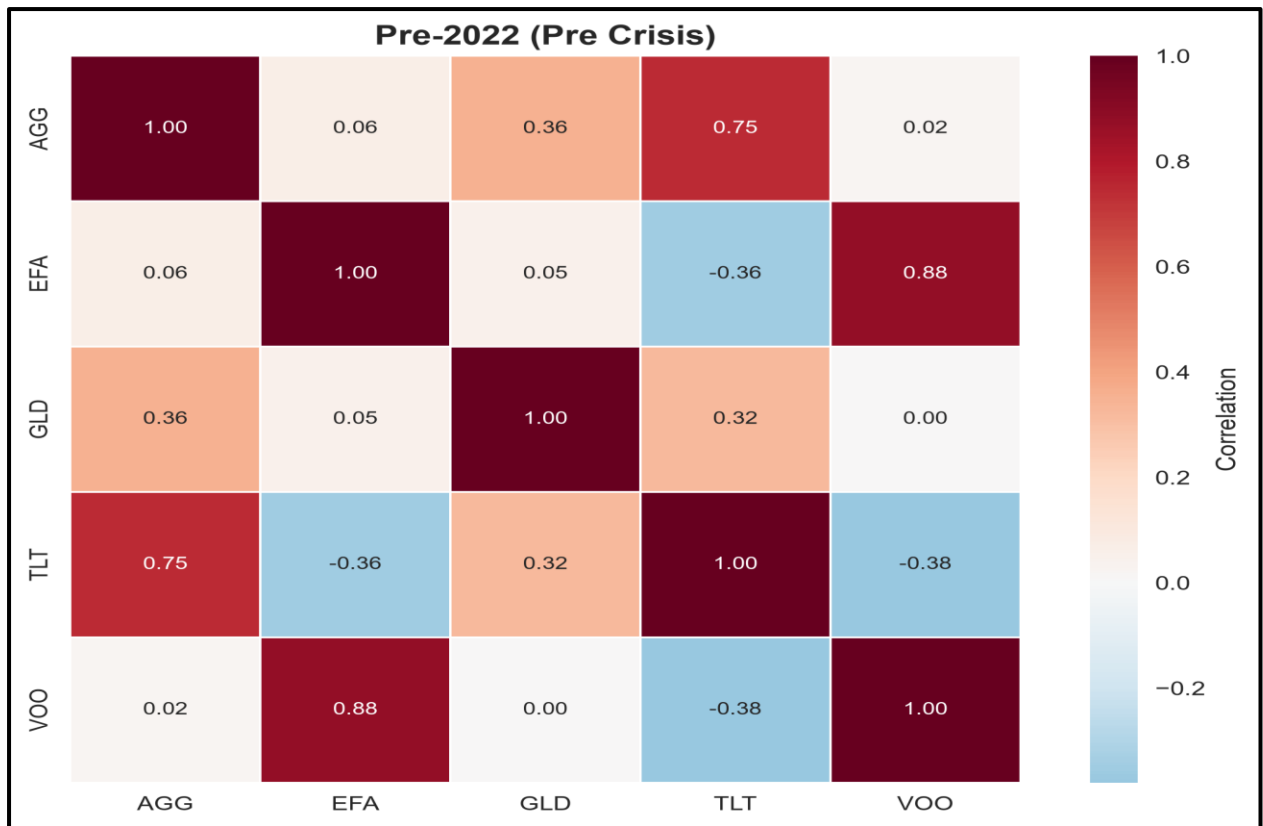


Figure 5.4

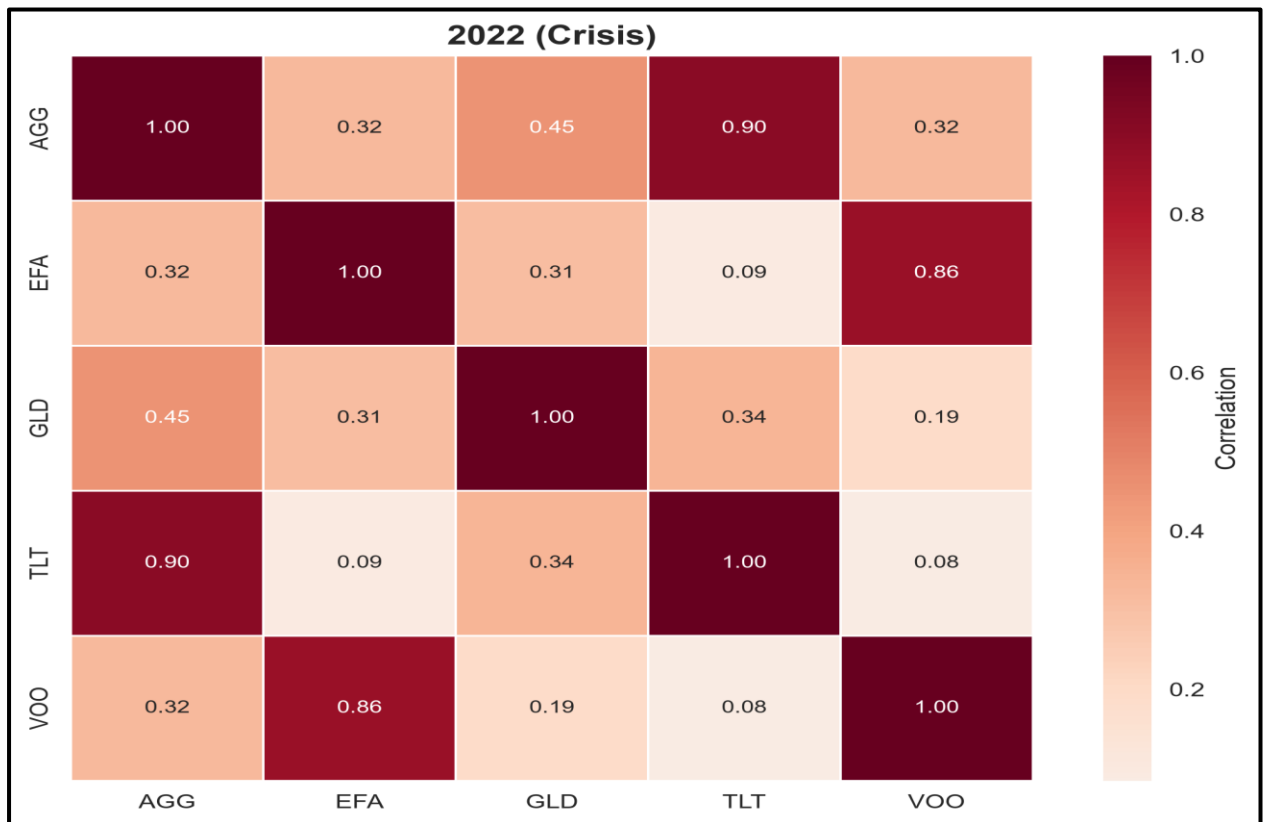


Figure 5.5

The most striking change is between **VOO** and **TLT**. Per **Figure 5.4**, the correlation prior to 2022 was -0.38. When equities fell, bonds tended to rise, providing a cushion. This negative correlation is the foundation of traditional balanced portfolios like the **Classic 60/40**.

In 2022, that relationship flipped. The correlation turned positive, rising to +0.08 as seen in **Figure 5.5**. Equities and bonds fell together, meaning the diversification benefit disappeared exactly when it was needed most. The change of +0.46 is substantial and explains why the backtest results in 2022 were so poor for bond-heavy portfolios.

This correlation breakdown also helps explain the stress test results. The **Defensive** portfolio performed well in the **2008 Crisis** scenario, when falling equities were offset by rising bonds, but suffered a 19% loss under the **Solvency II** scenario where all assets declined simultaneously. This demonstrates that portfolio diversification can provide limited protection when correlations increase across asset classes.

Overall, the results show that tail risk is dynamic and can increase when historical relationships between assets break down. Combining heavy-tailed models, backtesting and scenario analysis therefore provides a more robust assessment of portfolio risk than relying solely on historical relationships.

Dashboard

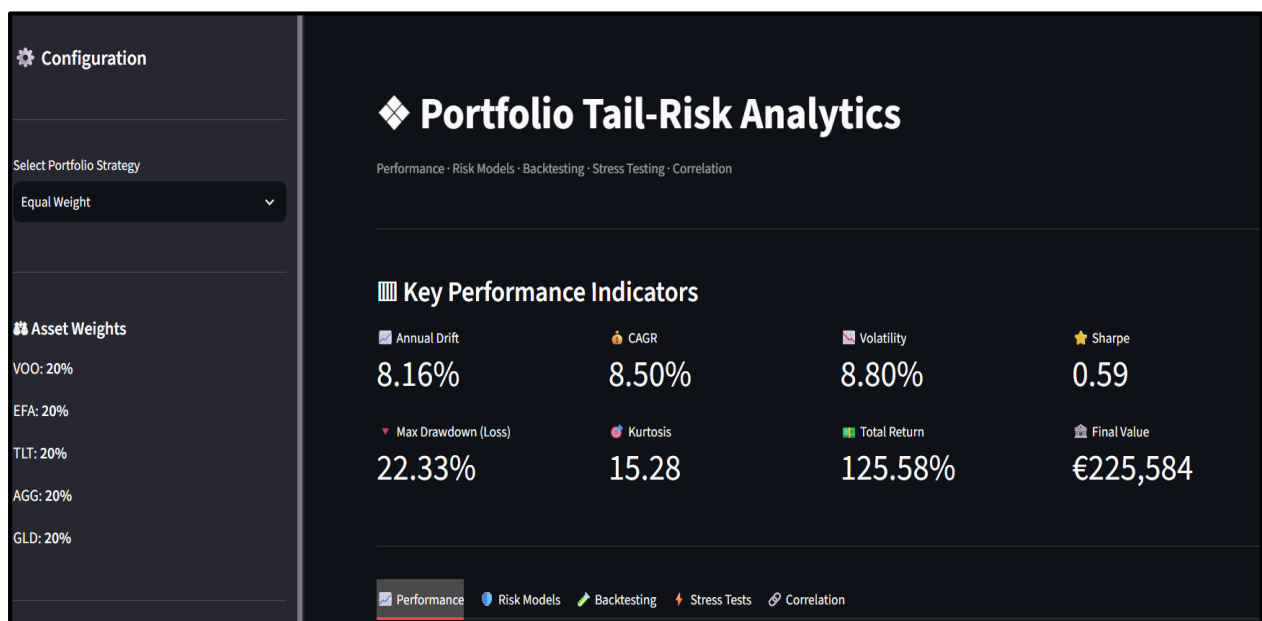
6.1 Dashboard Summary

In addition to the static analysis phases, an interactive dashboard was built bringing together the key results from Phases 01–07 into a single, user-friendly interface.

The dashboard allows users to:

- Select any of the five portfolio strategies.
- Adjust the initial investment amount (€10k – €1m).
- Explore performance metrics, VaR/ES estimates, backtest results, stress test losses, and correlation changes through dedicated tabs.
- Switch between different risk models and scenarios instantly.

Figure 6.1 below provides a preview of the main interface.



Conclusion

7.1 Conclusion

The project interpreted how various risk models assess the tail risk of a portfolio and estimate severe losses. It used daily data from the period 2016 to 2025 for five multi-asset portfolios and compared the Historical Simulation, Normal, Student-t and Bootstrap methods in terms of VaR and Expected Shortfall, carrying out expanding-window backtesting and stress testing.

The findings support the main hypothesis; the Normal model consistently failed to adequately account for tail risk and had the greatest number of VaR breaches during the 2022 stress period. The Student-t model gave more conservative estimates and delivered the best results at the 99% level for the **Equal Weight** portfolio in 2022, whereas Monte Carlo Bootstrap was very similar to Historical Simulation.

The analysis also showed that tail risk increases when market relationships change. In 2022, the breakdown of the traditional negative equity-bond correlation reduced diversification benefits, while stress scenarios involving simultaneous asset losses produced the largest portfolio losses. Overall, the findings support combining heavy-tailed models, historical simulation, backtesting and stress testing for a more robust assessment of portfolio risk.

7.2 Limitations and Future Work

The study is primarily concerned with daily, one-day-ahead risk and makes use of deterministic stress scenarios. The Monte Carlo Bootstrap method also fails to explicitly take time dependence or volatility clustering into account. In future research, it may be possible to include GARCH, Extreme Value Theory, filtered historical simulation and more sophisticated dependence modelling, as well as to consider longer risk horizons and additional stress periods.

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- Yahoo Finance. (2026). *Historical Adjusted Closing Prices for VOO, EFA, TLT, AGG, and GLD (Jan 2016 – Dec 2025)*. Available at: <https://finance.yahoo.com>